Digital Signal Processing

Course Code: EC305 · Department: Electronics & Communication Engineering · Semester 5

Course Overview

This course introduces the fundamental concepts of digital signal processing, including discrete-time signals and systems, the z-transform, Fourier analysis, and digital filter design. Students will learn to analyze signals in both the time and frequency domains and implement practical DSP algorithms. The course emphasizes both theoretical foundations and hands-on implementation using Python and MATLAB. By the end of the course, students should be able to design finite impulse response and infinite impulse response filters for real-world applications such as audio processing, image enhancement, and communication systems.

Learning Outcomes

- Analyze discrete-time signals and linear time-invariant systems.
- Compute the z-transform and its inverse for common signal sequences.
- Apply the discrete Fourier transform and the fast Fourier transform algorithm.
- Design and implement FIR and IIR digital filters using windowing and bilinear transform methods.
- Evaluate the effects of sampling, aliasing, and quantization on signal fidelity.

Prerequisites

Signals and Systems (EC204), Linear Algebra (MA201), and basic programming proficiency. The sampling theorem states that a continuous signal must be sampled at a rate greater than twice its highest frequency component to avoid aliasing; this fundamental bound is known as the Nyquist rate.

Unit 1: Discrete-Time Signals and Systems

Representation of discrete-time signals, basic operations on sequences, classification of systems as linear or nonlinear, time-invariant or time-varying, causal or non-causal, and stable or unstable. Convolution sum and its properties. Difference equations and their solution using iterative and closed-form methods.

Unit 2: The z-Transform

Definition and region of convergence, properties of the z-transform, inverse z-transform via partial fraction expansion and power series. Transfer functions of linear time-invariant systems, poles and zeros, and their relationship to system stability and frequency response.

Unit 3: Fourier Analysis of Discrete-Time Signals

The discrete-time Fourier transform, the discrete Fourier transform and its properties, circular convolution, linear convolution using the DFT, and the fast Fourier transform algorithm with radix-2 decimation-in-time and decimation-in-frequency structures. Applications of the FFT in spectral analysis and convolution of long sequences.

Unit 4: Digital Filter Design

Design of FIR filters using the window method and frequency sampling, design of IIR filters from analog prototypes using impulse invariance and the bilinear transform, filter structures including direct form, cascade form, and parallel form, and finite word-length effects including quantization error and coefficient sensitivity.

Unit 5: Applications and Case Studies

Speech signal processing fundamentals, audio effects including echo and reverb, image filtering using 2D convolution, and an introduction to adaptive filters with the LMS algorithm. A case study on removing noise from audio recordings using a band-pass FIR filter designed in Python.

Assessment Scheme

Component	Weight	Details
Mid-Semester Exam	30%	Units 1–2, closed book
End-Semester Exam	50%	All units, closed book
Assignments	10%	4 programming assignments in Python
Lab Work	10%	MATLAB/Octave experiments

Recommended Textbooks

1. Oppenheim & Schafer, "Discrete-Time Signal Processing", 3rd Edition, Pearson.
2. Proakis & Manolakis, "Digital Signal Processing: Principles, Algorithms and Applications", 4th Edition, Pearson.
3. Mitra, "Digital Signal Processing: A Computer-Based Approach", 4th Edition, McGraw-Hill.

Faculty

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